

Shubham Kumar

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EDUCATION

ITM University

B.Tech in Computer Science and Engineering

CGPA: 8.3/10.0

2023 – 2027

TECHNICAL SKILLS

Languages:	C, C++, Java, Python, SQL
Core CS:	Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems
Backend / Systems:	File Handling, MySQL, REST APIs, JWT Authentication
Frameworks / Libraries:	NumPy, Pandas, Scikit-learn, YOLOv8, OpenCV
Deployment / Tools:	Vercel, Postman, Git, Vite
Web:	HTML, CSS, JavaScript, React

RELEVANT COURSEWORK

Data Structures & Algorithms · Object-Oriented Programming · Database Management Systems · Operating Systems · Computer Networks · Software Engineering · Discrete Mathematics · Machine Learning

EXPERIENCE

Software Development Intern — Blog Post Management System

Intellige IT Studio

Industrial Training

Java, MySQL, REST APIs

- Completed industry-oriented software development training and built a fully functional blog post management system using **Java, MySQL, HTML, CSS, REST APIs, and Postman**.
- Designed and implemented **REST**-based backend APIs for blog and user management supporting full CRUD operations (create, update, search, delete).
- Implemented user and admin roles with role-based access control for secure content management.
- Enabled tracking of creation and update timestamps for blogs, ensuring data audit capabilities.

PROJECTS

MCP Server — AI-Driven 3D Model Generation | Python, Socket Programming, Blender, Cursor AI

- Designed and implemented a custom **MCP server** using socket-based communication for real-time tool integration.
- Integrated **Blender** and **Cursor AI** to enable 3D object generation using natural language prompts.
- Built a scalable, real-time system capable of connecting multiple tools simultaneously.
- Enabled full customization and modification of 3D models based on user input.
- Implemented robust error handling and reconnection logic to maintain stable socket connections during long-running generation tasks.
- Structured the MCP protocol interface to be extensible, enabling future AI tools to plug into the server with minimal integration overhead.

CivicChain — Civic Intelligence Platform | React, TypeScript, Vite, Node.js, Express, PostgreSQL, Prisma

- Built a full-stack civic-tech web application enabling citizens to report geo-tagged civic issues, track resolution status, and monitor ward-level official accountability and public spending.
- Architected the frontend with **React, TypeScript, and Vite**, integrating **Mapbox GL JS** for interactive WebGL mapping with heatmaps, 3D buildings, and custom geographic clustering.
- Designed a **PostgreSQL** database with **Prisma ORM** to model issues, ward officials, and users with relational mappings, replacing the initial **MongoDB** prototype for stronger relational integrity.
- Implemented secure **JWT**-based authentication with role-based access and a dual-mode auth system supporting both local and API-based sign-in.
- Developed a **REST API** using **Express** and **Node.js** with endpoints for issue reporting, community upvoting, and a ward-official accountability dashboard (NetaMeter) with trust and resolution scores.
- Deployed the application on **Vercel** with a live production build, using **Neon serverless PostgreSQL** for the database layer.

MINI PROJECTS

- **To-Do Application** — Full-stack task management app with CRUD functionality.
- **Chatbot** — Interactive conversational bot using rule-based/NLP techniques.
- **Calculator** — Multi-function calculator application with scientific operations.
- **Profile Calculator** — Profile analytics and scoring tool.
- **Number Guessing Game** — Interactive CLI game with randomized logic and difficulty levels.

ACHIEVEMENTS & HACKATHONS

- **Winner** — Hackathon conducted by ITM University.
- **Finalist** — “FinShield” Hackathon conducted by Bank of India; presented solution at **IIT Hyderabad**.
- **Participant** — Smart India Hackathon (SIH); actively participated in multiple inter-college hackathons.